System Architecture Explanation

This document explains the system architecture for the **Crowdsourced Civic Issue Reporting and Resolution System**. The architecture has been designed to ensure transparency, scalability, and user engagement while involving both citizens and government officials.

1. Citizens (Users)
Citizens interact through a **Mobile/Web Progressive Web App (PWA)**. The features include:
- Reporting issues with photo, location, and description/voice.
- Tracking status and updates of submitted issues.
- Providing confirmation when issues are resolved.
- Escalating issues if not resolved within time.
- Earning rewards through gamification for active participation.

2. Backend System (Node.js + Express)
The backend is the core engine, responsible for managing API calls, authentication, notifications, rewards, and escalations. It ensures smooth communication between the user app, database, and admin dashboard.

Main components:
- API Gateway: Handles all user/admin requests.
- Authentication: Managed with Clerk / Firebase Auth.
- Issue Management Service: Stores and processes reported issues.
- Notification Service: Sends SMS, Email, and Push notifications.
- Gamification Engine: Manages points and rewards.
- Escalation Handler: Routes unresolved issues to higher officials.
- Analytics Engine: Provides insights on civic engagement and resolution trends.

3. AI Module (ML/NLP)
The AI module enhances efficiency by:
- Detecting duplicate reports.
- Predicting priority/urgency using text and image data.
- Automatically routing issues to the correct department.
- Generating insights and reports for better decision-making.

4. Database Layer
The system uses a scalable database (e.g., MongoDB / PostgreSQL) with key collections/tables:
- Users
- Civic Issues
- Departments
- Rewards & Points
- Feedback & Logs
- Escalation Records

5. Govt. Officials (Admin Dashboard)
The admin dashboard is for government officials and department staff. It provides:
- Secure Login for staff.
- Viewing and filtering issues.
- Assigning issues to relevant departments.
- Updating progress and resolution status.
- Viewing escalations and acting upon them.
- Access to analytics and performance reports.

6. Data Flow Summary

1. Citizen reports an issue via the app.
2. Backend processes the request, authenticates the user, and stores the issue in the database.
3. AI module checks for duplicates, sets priority, and routes to the correct department.
4. Admins view and act on issues via the dashboard.
5. Citizens receive notifications on progress.
6. Once resolved, citizens confirm closure and provide feedback.
7. Gamification rewards are assigned based on user participation.
8. Escalation handler ensures unresolved issues are flagged to higher authorities.